

EvoRank: LLM-Guided Evolution of Multi-Objective Learning-to-Rank Pipelines^{*}

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Abstract

We present EvoRank, an open autonomous ranking engineer: an LLM-guided evolutionary loop that discovers complete Learning-to-Rank pipelines (features, models, losses, ensembles) for multi-objective e-commerce search. On the Expedia ICDM 2013 dataset, with relevance, conversion, and revenue as competing objectives, three independent runs each converge within 50 iterations (about ten dollars) on interpretable pipelines that beat an Optuna-tuned LambdaMART on 60k held-out queries, an advantage that persists at full data scale and places in the top 6 percent of the original competition. A first campaign, evolving only training objectives, builds the central design rule: it appeared to work on its selection fold (the small dataset it uses to pick winners) while a transfer audit, re-scoring winners on held-out data, showed the gains were almost entirely fitness noise (the randomness of its own scoring), and neither seeded domain knowledge nor richer diagnostic feedback changed what transferred. The deciding quantity is measurable in advance: search-space headroom relative to fitness noise. We package this as a headroom gate that predicts, before any LLM spend, whether the loop will pay off, and we release the system, the auditing tools, and a catalog of failure modes with their guardrails, so teams can apply the procedure to their own ranking stacks.

Keywords

learning to rank, LLM agents, automated machine learning, evolutionary program search, e-commerce search, reproducibility

1. Introduction

Automated discovery loops that pair a large language model with evolutionary search over programs have produced striking results, and industrial systems such as Meta’s Ranking Engineer Agent apply the pattern to ranking at scale. These systems are closed, and published results in the genre share a gap: gains are overwhelmingly reported on the data the loop used to select candidates. For ranking, where per-candidate evaluations are noisy by necessity, this matters: selection under noise promotes lucky candidates, and a loop can appear to make steady progress while discovering nothing that transfers.

We contribute an open, fully audited account of when the pattern works and when it silently does not, on a public e-commerce ranking task with three competing objectives (relevance, conversion, revenue). Our contributions are: (1) a quantified boundary condition for when such loops work: evolved training objectives whose selection-fold gains do not survive a held-out transfer audit, because real effect sizes are smaller than the noise of the loop’s own scoring; (2) controlled ablations of guidance showing that seeded domain knowledge and evidence-rich diagnostic feedback change the speed and style of search but not what generalizes; (3) the positive case under identical auditing: pointing the same loop at a search space whose effect sizes clear the noise floor yields transfer-audited pipelines, better on every objective at once, in all three runs, which reconstruct and extend the ICDM 2013 challenge winners’ hand-built solutions; and (4) the released tool and its audit stack (the acceptance and stopping criteria

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^{*}Code, configurations, seeds, discovered programs, and all machine-generated results are available at <https://github.com/shabazpatel/evorank>.

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applied to every result), including the search-space headroom gate, a cheap pre-check of whether a search space will reward the loop.

2. Related Work

LLM-guided program evolution. FunSearch [1] and AlphaEvolve [2] pair an LLM mutation operator with evolutionary search over programs and automated evaluators, with open engines [3, 4] reproducing the pattern; industrial ranking agents (REA, GEARS) [5] apply agentic experimentation to production ranking behind closed doors. We add the missing measurement layer: the same loop shown selecting noise in one search space and discovering in another, in an open, reproducible instance of the pattern for tabular LTR.

Automated loss discovery. GLO [6], AM-LFS [7], AutoLoss-Zero [8], and CSE-AutoLoss [9] search loss functions from small mathematical building blocks, mostly for vision; LambdaLoss [10] is the hand-derived counterpart (a loss built from a bound on the ranking metric); RankEvolve [11] applies evolutionary search to unsupervised lexical retrieval with a single-number score. We bring loss search to supervised multi-objective ranking with gradient-boosted trees (GBDT) via LLM mutations over full gradient code, and map a boundary this literature has not measured: the family’s headroom sits below the noise floor of any evaluation fast enough to run inside a loop, so discovered losses do not transfer.

Reflective feedback for LLM optimization. Reflexion [12], ProTeGi [13], OPRO [14], and GEPA [15] show that richer textual feedback (traces, textual gradients) improves LLM-driven search. We measure that assumption’s boundary: feedback that inspects the candidate’s own training gradients (possible because the evolved program is a loss function) changed nothing about what transferred across three runs, while simply enlarging the evaluation fold tripled it.

Agentic AutoML and LLM feature engineering. CAAFE [16] generates features for tabular classification; AIDE [17], ML-Agent [18], and related agents iterate over ML pipelines, typically evaluated by single-objective validation accuracy. We differ in the ranking domain (query-grouped data, per-query features and ensembling), in transfer-audited selection as the acceptance bar, and in a quantified catalog of target-encoding leakage modes.

LTR foundations and the benchmark. LambdaMART [19] remains the standard for feature-based ranking, where gradient-boosted trees still match or beat neural rankers [20]; the ICDM 2013 challenge solutions [21] document the winning feature recipes and score-blending ensembles; a 2024 survey [22] notes that public datasets with monetary outcome labels remain scarce, anchoring open multi-objective work to this benchmark. Our loop rediscovers and extends the essential structure of those solutions autonomously, for about ten dollars.

3. System and Audit Methodology

3.1. Background: the evolutionary engine

EvoRank builds on SkyDiscover, an open framework for LLM-guided evolutionary program search in the AlphaEvolve family [2, 3] (Figure 1). The unit of evolution is a source file in which mutable regions are delimited by EVOLVE-BLOCK markers; everything outside the markers is frozen. Each iteration samples a parent (plus context programs) from a program database, assembles a prompt with their source, scores, and feedback, makes one LLM call that edits the mutable regions, evaluates the candidate with a user-supplied evaluate function, and inserts it back into the database.

We use SkyDiscover’s AdaEvolve search controller [3]. Its program database keeps several semi-isolated subpopulations (islands; two here, spawning up to five on stagnation) that periodically exchange their best programs; a bandit allocates iterations to islands by recent improvement, and each island adapts its exploration intensity accordingly. The database keeps a Pareto archive, the set of best available trade-offs over the three objectives (ndcg, booking ndcg, pooled dollar-weighted revenue), with a single-number score used only for tie-breaking.

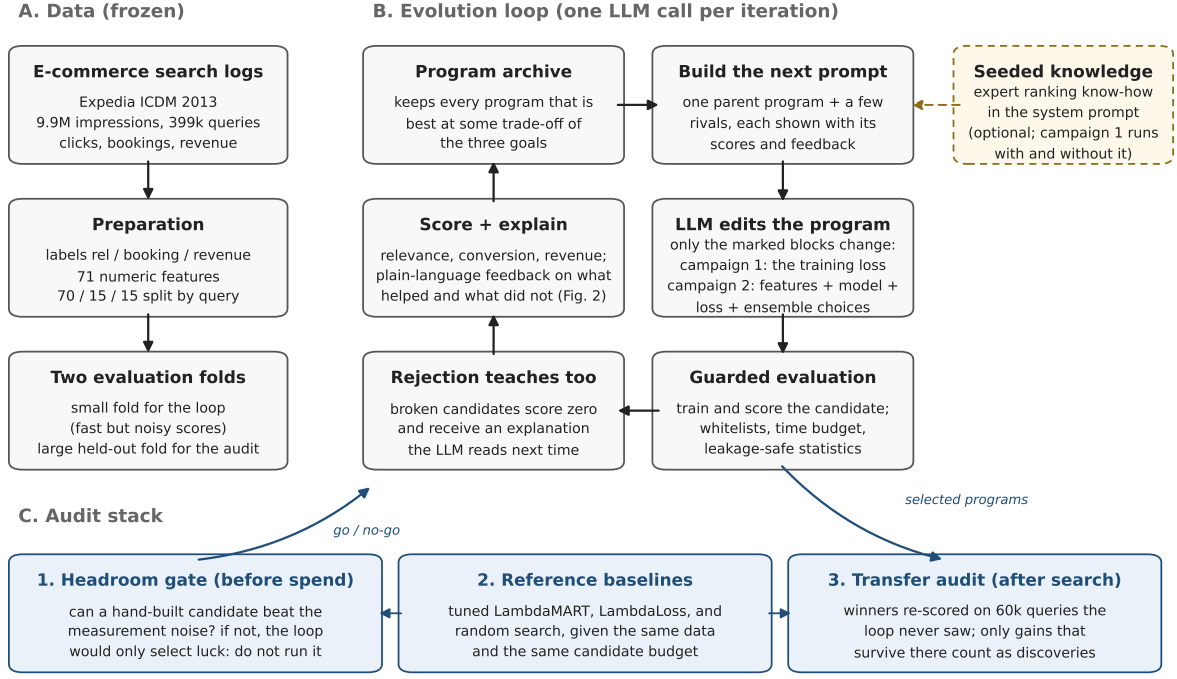


Figure 1: The EvoRank system. Lane A: frozen data preparation producing a small scoring fold and a large held-out fold. Lane B: the loop, one LLM call per iteration, from archive through prompt, edit, guarded evaluation, and feedback (Figure 2) back to the archive. Dashed: optional expert knowledge, ablated in campaign 1. Lane C: the audit stack: pre-spend headroom gate, baselines on equal terms, and the held-out audit that decides what counts.

3.2. Preliminaries: the algorithms in brief

Ranking and NDCG@k. A query returns a list of items with relevance labels. NDCG@k scores how well a model orders each list: relevant items near the top earn gain discounted by position, normalized so 1.0 is perfect. We average it per query (for revenue we instead pool dollars, so high-value queries weigh more).

GBDT and LambdaMART. Gradient-boosted decision trees (XGBoost [23], LightGBM [24]) fit a sequence of small trees, each correcting the previous ones via gradients of a loss. LambdaMART is the ranking instance: for every within-query pair where one item outranks another in the labels, it applies a pairwise gradient scaled by how much swapping the two would change NDCG, concentrating training pressure where the metric is at stake. XGBoost and LightGBM implement it; campaign 1 evolves exactly this gradient function.

Hyperparameter tuning with Optuna. A GBDT’s quality depends on learning rate, depth, subsampling, and regularization. Optuna [25] searches that space with Bayesian trials scored on validation data. Baselines get 40 trials, re-run at every training-set size we report, since optimal settings change with data volume.

Ensembling by per-query z-scores. Different model families make partially independent errors; because their scores live on different scales, each model’s scores are standardized within each query before a weighted sum. It was the challenge winners’ choice and what our pipelines select.

Out-of-fold target encoding. Encoding a categorical column by the average label of its value (a hotel’s historical booking rate) leaks labels into features unless each row is encoded from other query-folds only.

The statistics. A paired query bootstrap resamples queries and recomputes the difference between two systems many times (10,000 in the audit, a few hundred inside the loop’s noise gate), treating the query as the unit of noise. Pareto dominance means at least as good on every objective and better on

one; hypervolume measures how much objective space the non-dominated set covers.

3.3. Task, data, and evaluators

Data is the public Expedia Personalized Hotel Search training file (9.9M impressions, 399k queries), split 70/15/15 by query. Each shown hotel gets a graded relevance label: 5 if booked, 1 if clicked, 0 otherwise. The loop evaluates candidates on a small fitness fold of 8k training queries: a candidate trains a gradient-boosted ranker under a fixed budget and is scored on the fold's held-back validation queries; that score is its fitness, and the small fold keeps each evaluation between 5 and 40 seconds. Some candidates are broken rather than weak, for example gradients that turn infinite, training that exceeds the time budget, or a malformed pipeline specification. These receive zero fitness together with a message explaining what went wrong, which enters the next prompt, so the guardrails double as teaching signals.

3.4. The audit stack: acceptance and stopping criteria for the loop

The audit stack, applied identically to every campaign: (a) the headroom gate, run before any LLM spend: one hand-built candidate must beat the fitness fold's noise floor by a set margin, or the search space is not worth searching; (b) paired query bootstrap for every comparison, with noise-gated deltas inside the loop's own feedback; (c) equal-budget random-search controls; (d) equal tuning on both sides (discovered programs get the same Optuna budget as the baseline); (e) the transfer audit itself, re-scoring every checkpoint's selected program on 60k held-out queries; (f) three-objective frontier quality (hypervolume); and (g) component-removal tests on discovered programs. The tuned baseline throughout is LambdaMART (XGBoost rank:ndcg) with 40 Optuna trials.

4. Campaign 1: Objective Space, Where the Loop Selects Noise

The evolvable piece is the LambdaMART gradient function, computed per query group (the set of hotels returned for one search), with access to each impression's raw behavioral signals (click, booking, revenue for each shown hotel); training configuration, features, and metrics are frozen, and the starting version reproduces XGBoost's built-in rank:ndcg within noise.

On its selection fold, evolved objectives beat the default baseline and improve steadily. The transfer audit reverses the picture: final selections gain +0.0004 to +0.0014 NDCG@10 on held-out queries, the best evolved objective beats its own starting program by +0.0002 (two-sided $p = 0.76$), an equal-budget random search transfers as well or better ($p = 0.03$ in its favor), and every method is significantly below the tuned baseline ($p < 0.001$). In plain terms: on unseen data the evolved objectives are indistinguishable from where they started, blind mutation did as well as guided search, and ordinary tuning beat both. The mechanism is measured: component-removal tests put a single component's effect at 0.0005 to 0.008 NDCG@10, most below 0.005, while the selection fold's measurement noise (95 percent interval half-width) is ± 0.007 to 0.009 NDCG@10 and wider on booking and revenue, so picking the best score means picking luck. Enlarging the fitness fold's validation side from 1.6k to 6k queries halves that noise and triples the gain surviving transfer, the only intervention that moved it.

Guidance ablations: seeding the prompt with curated domain knowledge makes the search faster, not truer. Two of three seeded runs need only 1 to 3 iterations to match the best score an equal-budget random search reaches in its entire run (the third never does); unseeded runs need 7 to 23. Seeded runs also end with a better three-objective frontier in every run. What transfers to held-out data is unchanged. A diagnostic feedback channel (noise-gated deltas, per-segment breakdowns, probes into the candidate's training gradients) changes nothing measurable across three runs.

Analysis beat further search: removing the best objective's components one at a time (five evaluations in total) explained its behavior and exposed one component, a blend weight that adapted per query, as actively harmful; fixing that weight to a constant improved all three metrics. The family also

```
scores: ndcg=0.4357 (+0.0139 vs seed, +/-0.0054, SIGNIFICANT) |
       book_ndcg=0.4604 (+0.0142, SIGNIFICANT) | revenue=0.4309 (+0.0217, SIGNIFICANT)
members: [0] xgb/rank:ndcg: val ndcg 0.4325 (2s)
         [1] lgbm/lambdarank: val ndcg 0.4261 (4s)
         [2] sklearn/ert: val ndcg 0.4171 (28s)
ensemble: zscore_weighted [0.55, 0.33, 0.12] = 0.4357,
         best single 0.4325 (+0.0033)
```

Figure 2: Stage-attributed feedback for one campaign 2 candidate, excerpted from the run logs (feature-stage line elided). The +/- changes are measured against the starting pipeline and flagged SIGNIFICANT only above noise; per-member lines attribute quality to model and loss choices; the ensemble line reports the margin over the best member.

Table 1

Campaign 2 transfer audit: trained on the fitness fold, tested on 59,902 held-out queries. All runs beat the tuned baseline on NDCG@10; the best run (s2) beats it on all three objectives. Bold: best per column.

Method	Val NDCG@10	Test NDCG@10	Test book	Test revenue
Starting pipeline	0.4218	0.4222	0.4443	0.4105
LambdaMART + Optuna	0.4330	0.4296	0.4527	0.4231
Pipeline s0	0.4356	0.4316	0.4545	0.4257
Pipeline s1	0.4359	0.4322	0.4553	0.4215
Pipeline s2	0.4398	0.4352	0.4582	0.4239

flattens at scale: trained on the full dataset, every objective, including default rank:ndcg, lands within 0.001 NDCG@10 of every other.

5. Campaign 2: Pipeline Space, Where the Loop Discovers

Campaign 2 widens the search from the loss alone to the whole pipeline. The LLM now edits two blocks: feature-construction code over semantically named columns, where any statistic touching labels must come from helper functions with leakage guards built in (counts, out-of-fold rates, quantiles, and a train-fold linear score over query-level columns), and a pipeline specification from a whitelist (one to three members from XGBoost, LightGBM, or sklearn; ranking, pointwise, binary, or campaign-1 losses; z-score or rank-mean blending). Feedback is stage-attributed (each pipeline stage gets its own line) so the loop can tell which axis earned each change; Figure 2 shows an excerpt.

Before any LLM spend, we applied the headroom gate (Section 3.4): a hand-built candidate from the ICDM 2013 literature had to beat the noise floor by 2.5 times, and within-query rank features plus count encodings did (+0.013 NDCG@10). Building that candidate surfaced three successively subtler leakage bugs. Encoding a hotel by its booking rate on the training fold costs -0.10 NDCG: the model memorizes its own labels. The textbook fix, leaving each row’s own label out of its encoding, is worse at -0.13 : the exclusion shifts the encoded value in a direction that depends on the label, and the trees read that shift as the label itself. Only out-of-fold encodings, with folds split by query, are clean. The audit caught each bug before any search depended on it.

What the search found (Figure 3): three runs, at 50 iterations and ten dollars each, converged on the same interpretable core: within-query percentile ranks of price, star rating, review and location scores, count encodings of property and destination identifiers, a lean feature set (the programs’ own comments cite the earlier feedback showing that more features hurt), and a three-member ensemble of unlike models blended by per-query z-scores. One provenance caveat: the seeded knowledge included the gate’s finding that rank and count features carry headroom, so their presence is partly prompted; the specific constructions, model, loss, ensemble, and weight choices, and unprompted inventions (per-query price normalization, percentile-filling a mostly-missing location score) are the search’s own.

Does it transfer? Yes (Table 1): roughly 70 percent of selection-fold gains survive on 60k held-out

```
# discovered by run s1 (excerpt; ellipses and "# 5 columns" are editorial)
for c in ["price_usd", "prop_starrating", ...]: # 5 columns
    out[f"{c}_qrank"] = df.groupby("qid")[c].rank(pct=True)
for c in ["prop_id", "srch_destination_id"]:
    out[f"{c}_count"] = stats.count(df, c)

PIPELINE = {"models": [
    {"family": "xgb", "objective": "rank:ndcg", ...},
    {"family": "lgbm", "objective": "lambda:rank", ...},
    {"family": "sklearn", "objective": "ert", ...}],
    "ensemble": {"type": "zscore_weighted", "weights": [0.5, 0.35, 0.15]}}
```

Figure 3: The discovered program is readable code: feature excerpt and pipeline specification from run s1 (full programs in the repository).

Table 2

Full-scale comparison (280k train, 60k test queries), including NDCG@38, the challenge metric. The last row gives the pipeline’s XGBoost member the baseline’s full-scale tuned settings, scales its LightGBM member to match (255 leaves, 300 trees), and drops the third member; it had no search or tuning of its own.

Method	NDCG@10	NDCG@38	Book@10	Revenue@10
Tuned baseline (small-fold tuning)	0.4349	0.4983	0.4588	0.4300
Tuned baseline (full-scale tuning)	0.4544	0.5132	0.4805	0.4565
Pipeline s1 (small-fold tuning)	0.4490	0.5080	0.4738	0.4483
Pipeline s1 (baseline’s tuned settings, 2 members)	0.4619	0.5185	0.4883	0.4585

queries, and all three runs beat the tuned baseline there. A paired query bootstrap (10,000 resamples) makes this formal: the best run wins by +0.0057 NDCG@10 (95 percent CI [0.0041, 0.0073], one-sided $p < 0.0001$) and +0.0056 booking NDCG@10 ($p < 0.0001$), with revenue positive but within noise (+0.0008, CI [-0.0034, 0.0047]); the full-scale comparison shows the same pattern (+0.0075 NDCG@10, $p < 0.0001$). The multi-objective claim is therefore precise: significantly better relevance and conversion at matching revenue.

Full-dataset training yields its own methodology lesson (Table 2): against a baseline tuned on the small fold the pipelines win easily; re-tuning the baseline at full scale reverses that verdict; and only a comparison with comparable capacity on both sides isolates the discovered structure, which then wins on all four numbers. The margin splits across both components: the discovered features alone, under the same tuned single model, reach 0.4587 NDCG@10 (+0.0043); the two-member ensemble adds +0.0032 to 0.4619. We also scored both final models on the original competition’s hidden test set via Kaggle late submission (one shot each, no leaderboard iteration): the discovered pipeline reaches 0.5196 private NDCG@38, which would have placed 20th of 340 teams (top 6 percent) in the 2013 challenge, with the tuned baseline at 0.5140 (30th), so ten leaderboard places come from the discovered structure. The winning hand-built ensembles remain ahead (0.5398; our cited solution reports 0.5310), and our submission benefits from hindsight, since the winners’ published lessons are in the seeded knowledge.

6. Discussion

The two campaigns differ in exactly one variable, the search space, and the same loop produced noise in one and transfer-audited Pareto dominance in the other (Figure 4). The deciding quantity is measurable in advance: the ratio of achievable effect sizes to fitness-fold noise. We suggest the headroom gate as a cheap pre-spend go/no-go check for discovery loops, and transfer-audited selection as the reporting standard. Our results also rank where a team should invest next: evaluators first (everything that moved outcomes lived there: fold size, leakage-safe statistics, rejection messages the model learns from, the gate, and the audit); application spaces second (headroom relative to noise decides the outcome,

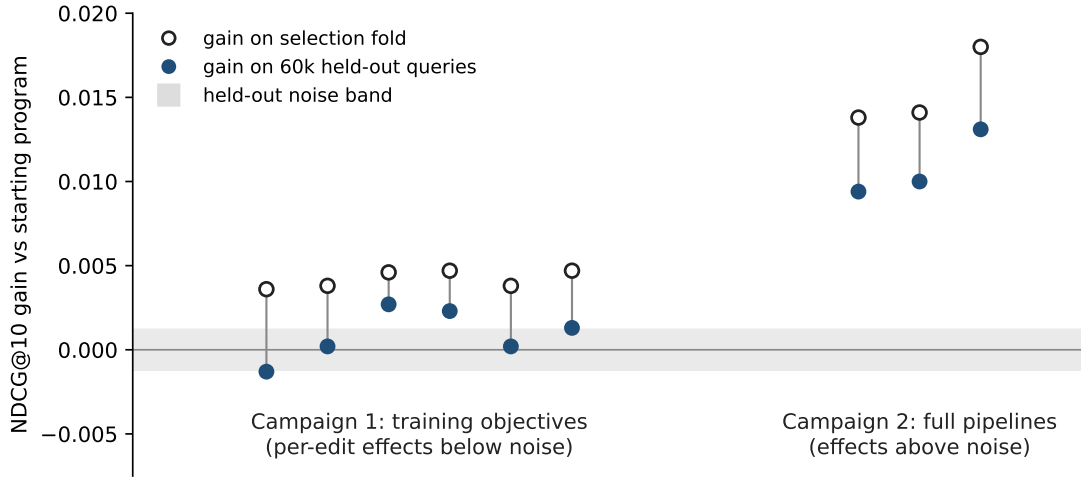


Figure 4: The study in one picture: per-run NDCG@10 gains over the starting program, on the selection fold (open circles) and on 60k held-out queries (filled). Campaign 1’s gains collapse into the noise band; campaign 2’s survive.

so choosing the space is choosing the result); search algorithms last (a recalibrated AdaEvolve and an EvoX meta-evolution variant [4], each at double budget, landed inside the spread of the original three runs, their internal signals drowned by the same noise). The natural next step is multi-fidelity evaluation: screen on the cheap fold, promote survivors to full scale before selection, and so expose the scale-dependent features (per-entity rates, position effects) behind much of our gap to the 2013 winners.

Limitations and generalization. Our evidence comes from one dataset, now thirteen years old, because newer public LTR datasets lack the conversion and revenue labels a multi-objective study needs [22]. The audit stack itself is dataset-agnostic: it needs only a small fitness fold, a larger held-out fold, a paired bootstrap on the metric of interest, and one strong tuned baseline, and the headroom gate is a recipe for any query-grouped ranking or recommendation task: measure the fitness fold’s noise by query bootstrap, hand-build one candidate from the domain’s literature, and require its gain to clear that noise by a set margin (2.5 times here). We compared two search spaces; intermediate ones (features plus loss, model plus loss) are untested, and the gate is the tool for ranking them before spending. Three runs per condition suffice for the paired-bootstrap headline, which every run cleared, but not to characterize the variance of the search itself. We did not compare against other LLM-guided discovery engines [2, 15]; the audit is engine-agnostic (the EvoX arm above is one such swap), and a controlled comparison is future work. Finally, revenue is a proxy metric, and discovered pipelines were audited for transfer, not for online effects.

7. Conclusion

For teams bringing LLM-driven discovery to their own ranking systems, this study distills to a working procedure: measure the noise of the evaluation you can afford; verify with one hand-built candidate that your search space offers effects above that noise; only then run the loop, with leakage-safe statistics and rejection messages the model can learn from; and let a held-out audit, not the loop’s own scores, decide what counts. Code, configs, seeds, logs, discovered programs, and audit scripts are released at <https://github.com/shabazpatel/evorank>.

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